

APQP FOR ENGINEERING & COMMODITY CHANGES

Scope Guidance for Supplier Changes, Material Changes, Process Changes | IATF 16949 Clause 8.3.6, 8.5.6

WHY APQP IS REQUIRED FOR CHANGES

IATF 16949 Clause 8.3.6 (Design and development changes) and Clause 8.5.6 (Control of changes) require that changes are assessed for impact and that appropriate APQP activities are conducted before implementation. Even seemingly minor changes can have significant quality implications.

CHANGE IMPACT ASSESSMENT MATRIX

CHANGE TYPE	EXAMPLE	PPAP REQUIRED?	APQP SCOPE	KEY RISK
New Part — New Design	Brand new part for new model	Yes — Level 3	Full APQP — All 5 phases	All risks apply
New Part — Carry-over Design	Same drawing, new customer	Yes — Level 2 or 3	Full APQP with LL from previous	Process feasibility
Supplier Change — Same Material	Changing steel supplier, same grade	Yes — Level 3	Phase 3, 4: New supplier PPAP	Material property variation
Material Change — Different Grade	Switching from cold-rolled to hot-rolled steel	Yes — Level 3	Phase 2, 3, 4: DFMEA update, PPAP	Design/Intentional Impact
Process Change — Same Plant	Switching welding machine, same process	Yes — Level 3	Phase 3, 4: PFMEA update, PTR	Process Capability
Process Change — New Plant	Moving production to different facility	Yes — Level 3	Full APQP Phases 3, 4, 5	All process risks
Tooling Change — Same Design	Replacing die, same part design	Yes — Level 2	Phase 3, 4: PFMEA, PTR, Cpk	Dimensional variation
Minor Engineering Change	Tolerance relaxation, cosmetic	Level 1 or 2	Impact assessment + Phase 4	Evidence of impact
Dormant Part (>12 months)	Part not produced for >1 year	Yes — Level 3	Phase 3, 4: Process reverification	Process drift
Sub-Supplier Change	Changing a raw material sub-supplier	Yes — Level 3	Phase 3, 4: Sub-supplier PPAP	Procurement quality

CHANGE NOTIFICATION REQUIREMENTS

NEVER implement a change without customer approval when the change affects:

- Form, fit, or function of the part
- Special (safety or significant) characteristics
- Any dimension or parameter included in the PPAP
- Manufacturing process, plant location, or production equipment
- Raw material supplier or material specification
- Subcontractors providing critical services (heat treat, plating, NDT)

CHANGE MANAGEMENT PROCESS FLOW

STEP	ACTION
1. Change Identification	Document the proposed change clearly — what, why, when
2. Impact Assessment	Assess impact on: design, process, SC, PPAP, customer
3. Risk Analysis	Update PFMEA with change. Assess risk level.
4. Customer Notification	Submit deviation/change request to customer if required
5. APQP Activities	Conduct appropriate APQP activities based on impact
6. PPAP Submission	Submit PPAP at appropriate level for customer approval

STEP	ACTION
7. Implementation	After customer approval — implement change
8. Verification	Verify change is effective — Cpk, quality metrics
9. Document Update	Update all affected documents: PFMEA, CP, WI, BOM, drawings
10. Lessons Learned	Capture lessons for future similar changes

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